PORTFOLIO CASE STUDY

ElbeFlow

Hamburg Urban Mobility Lakehouse

From five official SensorThings systems to a multilingual decision product

84,191	**~459M**	**102,994**	**2009 → live**
OFFICIAL STREAMS	SCHEDULED OBSERVATIONS	VERIFIED SAMPLE ROWS	HISTORICAL COVERAGE

PYTHON · PARQUET · DUCKDB · DBT · AIRFLOW · REACT · TYPESCRIPT

German by default · English · French · Arabic with right-to-left layout

ILLUSTRATED PROJECT REPORT · AUGUST 2026

A data platform that ends in a usable product

The project is designed to be evaluated at both engineering and product level.

01
INGEST
Paginated source adapters, bounded retries and incremental monthly windows.

02
STORE
Immutable JSONL/Gzip bronze and partitioned Parquet/ZSTD silver.

03
MODEL
DuckDB analytics, dbt transformations and explicit quality contracts.

04
DELIVER
Responsive React dashboard with live refresh and verified fallback.

Honest scale accounting

The application separates exact counts from estimates. 84,191 streams is a catalogue count. 102,994 rows is the exact committed sample count. ~459 million is a conservative coverage-by-cadence estimate for scheduled sources only; traffic-light and charging events are excluded.

Core reliability choices

- Idempotent partitions and half-open time windows
- Atomic completion, deduplication and SHA-256 verification
- Live API fallback without hiding freshness state
- Local-first design with a direct cloud migration path

Executive dashboard

German is the default experience for the Hamburg target market.

Scale in seconds

WHAT THE SCREEN PROVES

The opening view gives reviewers immediate evidence of the project's size, source diversity and reproducibility.

- 84,191 exact SensorThings streams
- ~459 million scheduled records
- 102,994 verified real sample rows
- Five official Hamburg data domains

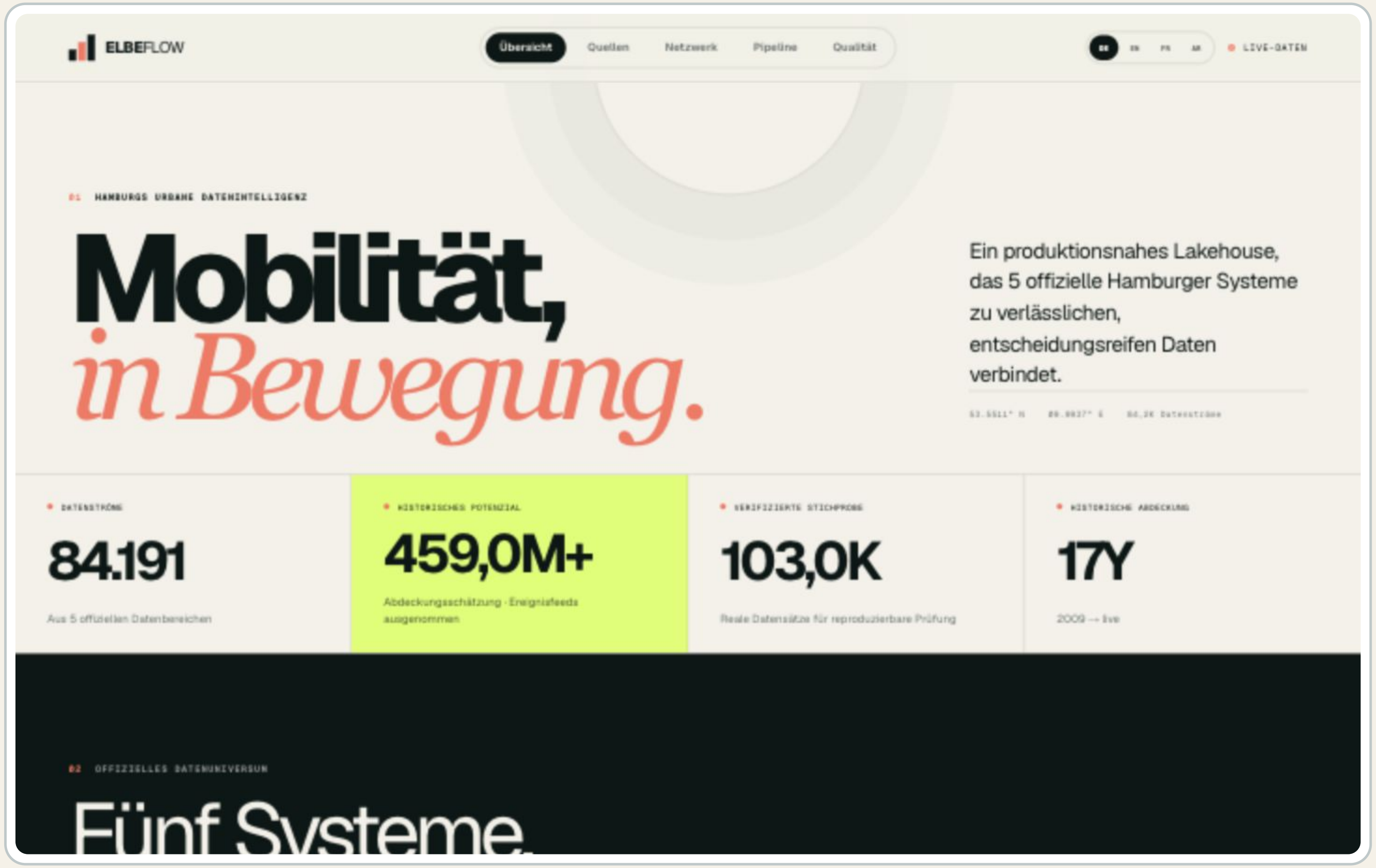

Figure 1 — Default German overview with live-status indicator and headline metrics.

Official source atlas

A single catalogue explains five very different operational datasets.

Source diversity

WHAT THE SCREEN PROVES

Each card communicates ownership, cadence, historical coverage, stream count and ingestion behavior.

- Traffic-light detector events from 2009
- EV charging events and site status
- 15-minute motor-traffic measurements
- 5-minute bicycle counts and StadtRAD availability

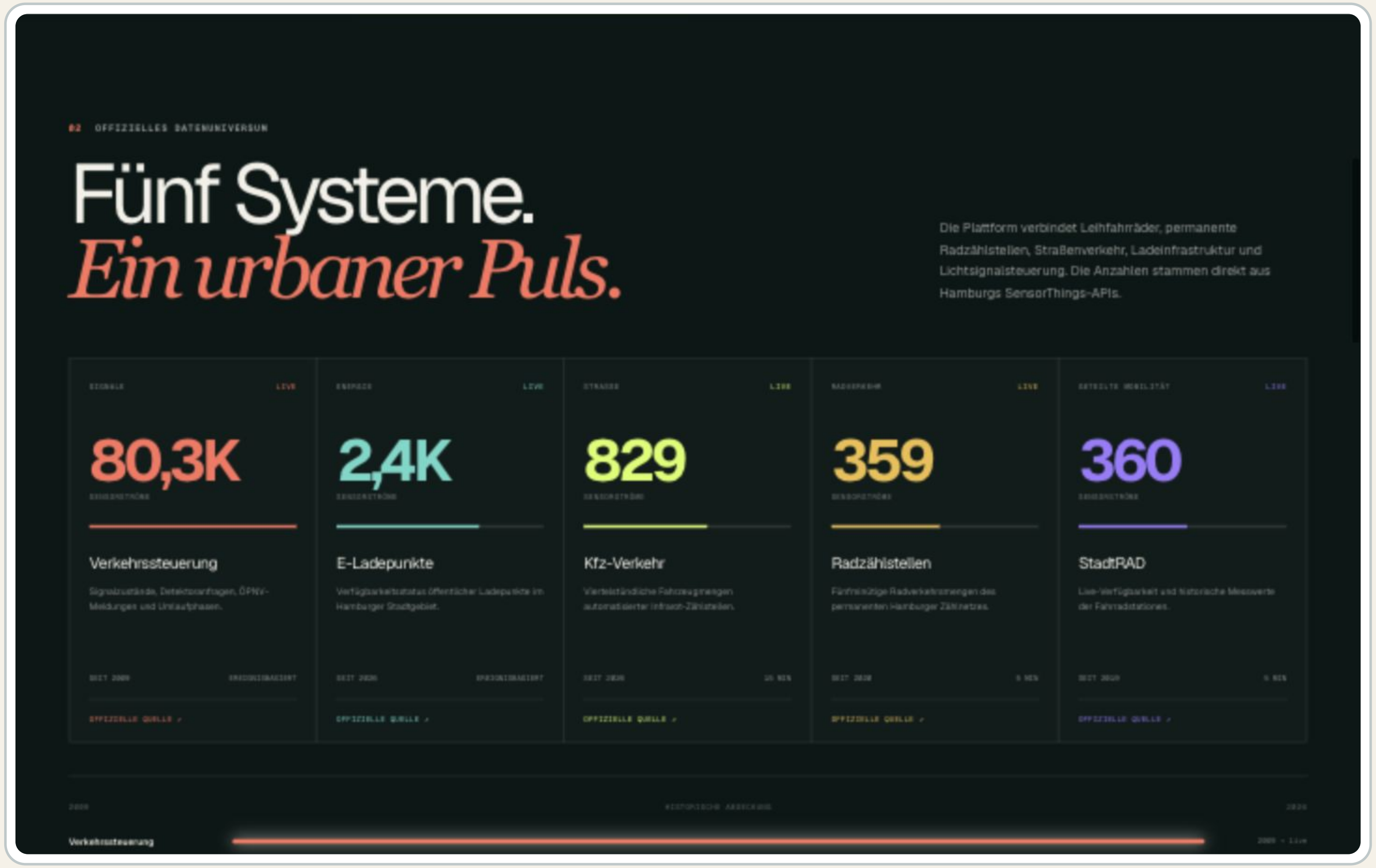

Figure 2 — Source catalogue with data-domain metadata and coverage boundaries.

Network intelligence

Spatial context and station performance turn raw availability records into decisions.

Operational value

WHAT THE SCREEN PROVES

The network view combines geography, current capacity, availability and station ranking in one analytical surface.

- Hamburg station locations and current status
- Capacity and bike-availability encoding
- Ranked station-level performance
- Snapshot fallback for resilient demonstrations

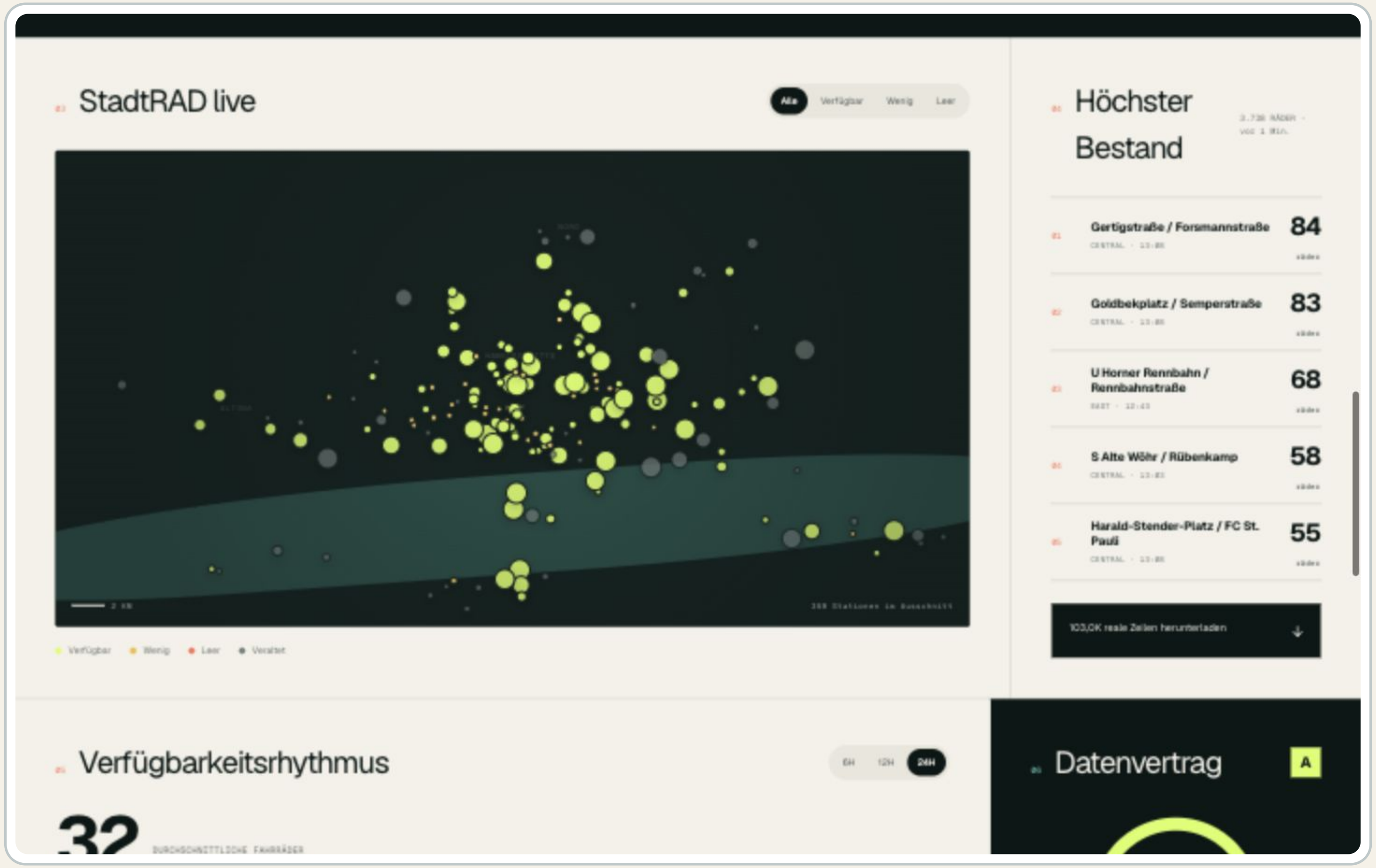

Figure 3 — Map and live station ranking built from official StadtRAD observations.

Quality and data contract

Reliability is presented as measurable product behavior.

Trust is visible

WHAT THE SCREEN PROVES

Freshness, completeness, validity and uniqueness are shown beside the rules that protect downstream analytics.

- Required-key and timestamp validation
- Duplicate observation protection
- Domain checks for impossible negative counts
- Clear separation of verified and estimated values

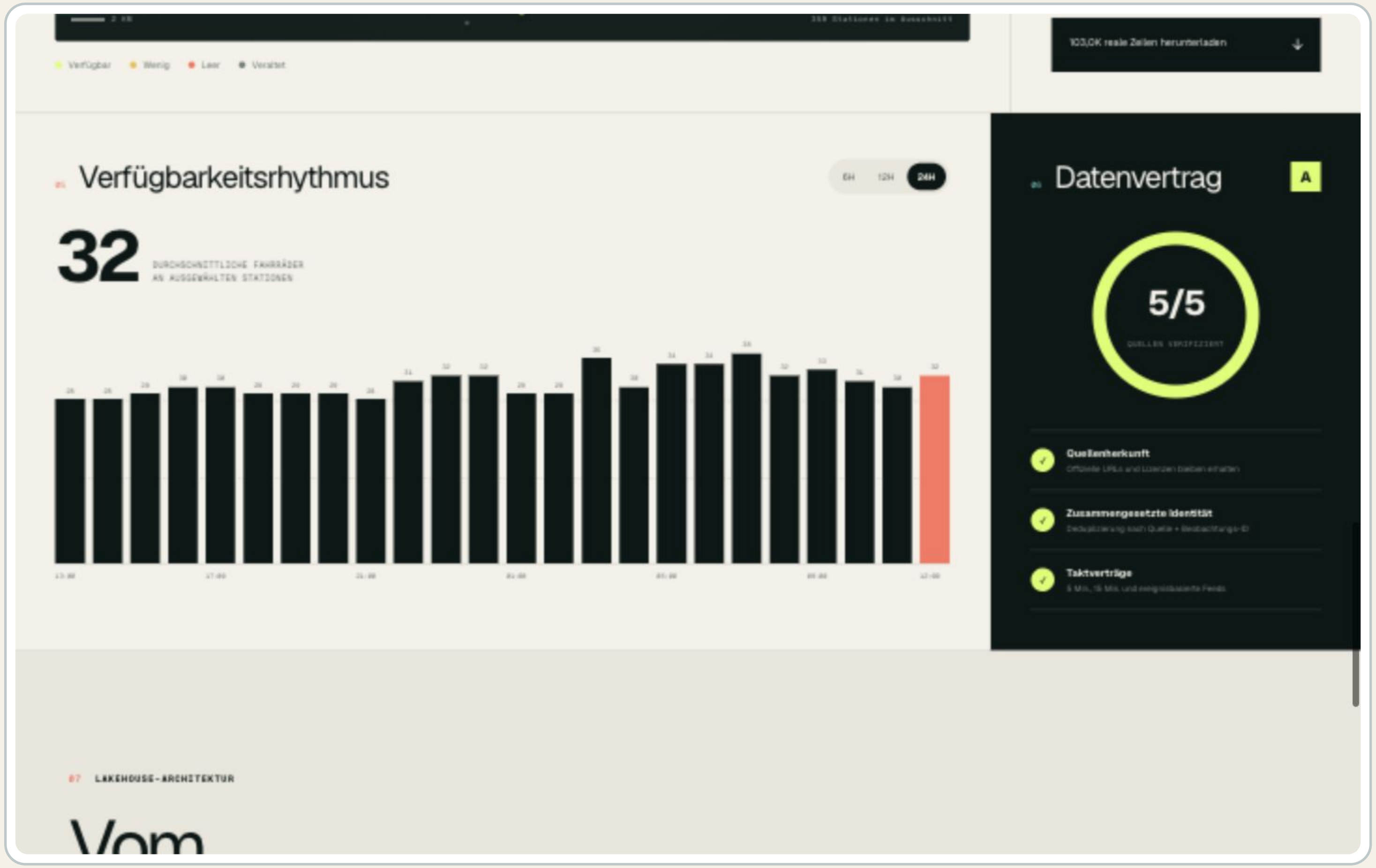

Figure 4 — Quality indicators and the explicit contract enforced by the pipeline.

Lakehouse architecture

The dashboard explains how the repository turns APIs into analytics.

End-to-end ownership

WHAT THE SCREEN PROVES

The architecture view connects ingestion, object storage, analytical modeling, orchestration and delivery.

- Immutable bronze for replayability
- Partitioned Parquet silver for efficient scans
- DuckDB + dbt for local analytics and contracts
- Airflow orchestration and automated validation

Figure 5 — Recruiter-readable bronze/silver/warehouse/product architecture.

English interface

The full product can be evaluated by an international hiring team.

International review

WHAT THE SCREEN PROVES

English localization covers navigation, source metadata, operational labels, quality explanations and formatting.

- Complete UI localization
- Locale-aware numeric and date formatting
- Stable analytical meaning across languages
- One-click language switching

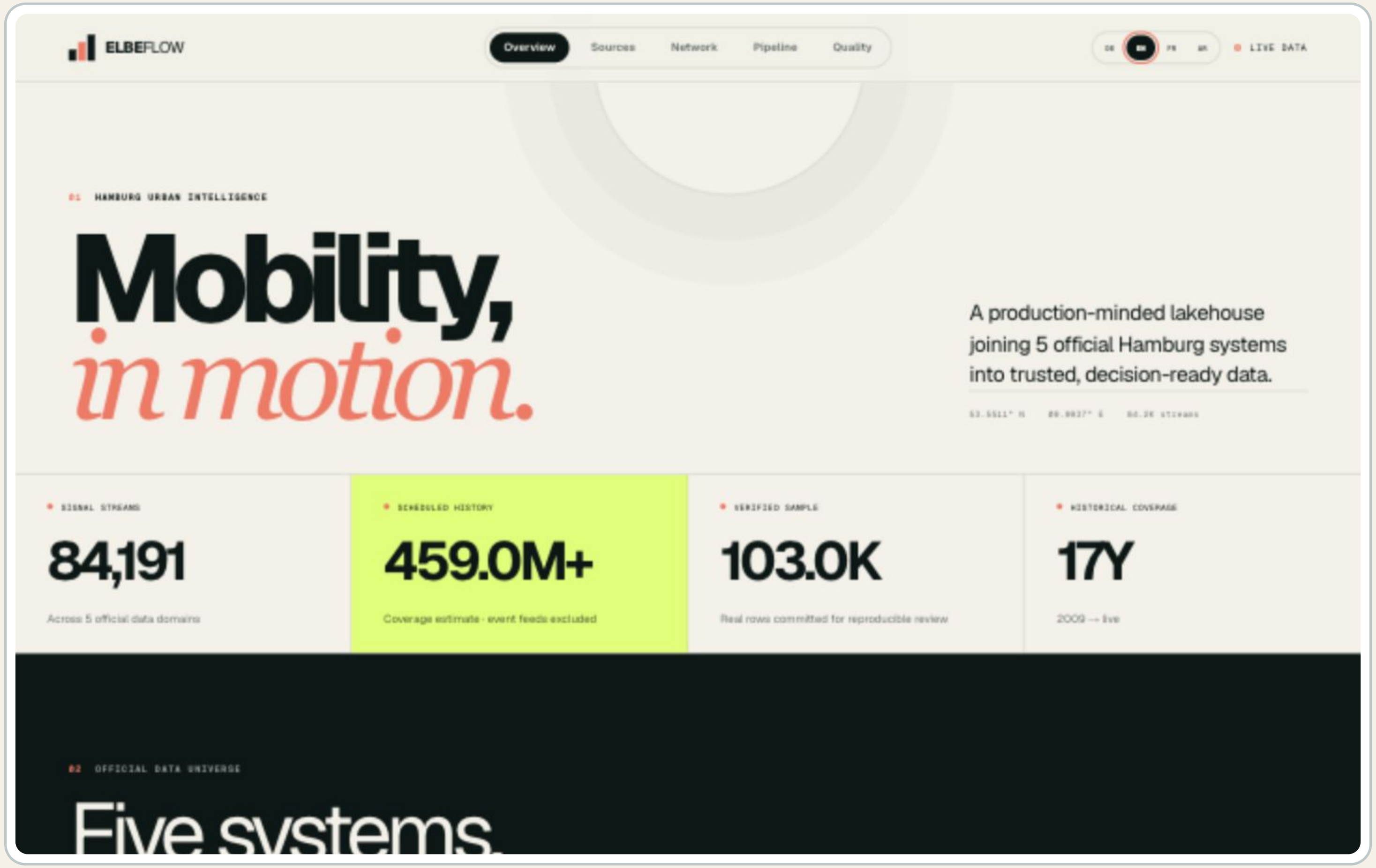

Figure 6 — English overview preserving the same analytical hierarchy as German.

French interface

Localization is implemented at the product layer, not as a static mock-up.

Consistent semantics

WHAT THE SCREEN PROVES

French users receive translated labels and explanations while every metric remains sourced from the same typed dataset.

- Translated product and engineering vocabulary
- French number formatting
- Identical data lineage and interactions
- Shared component system avoids interface drift

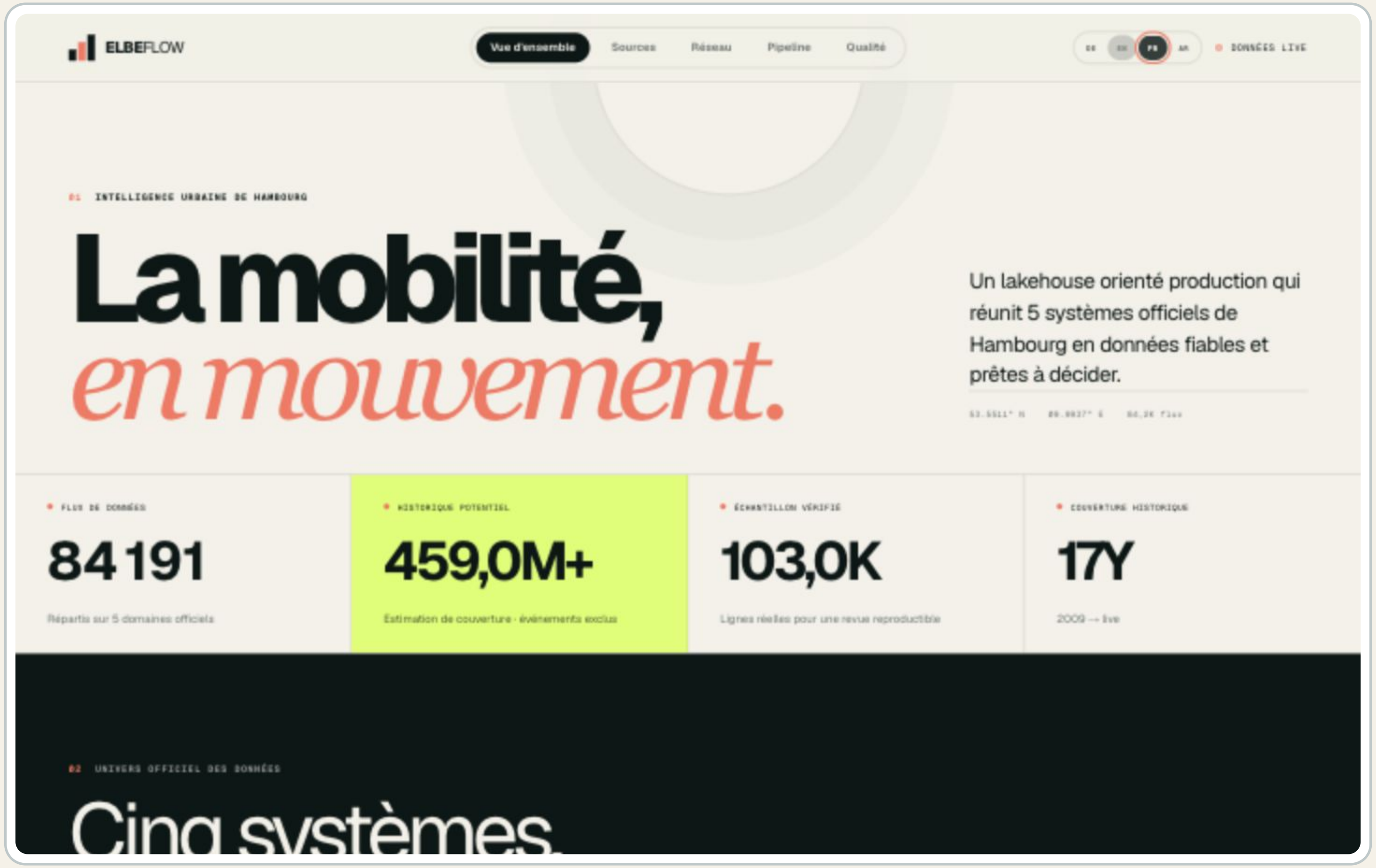

Figure 7 — French overview using the common data and component model.

Arabic right-to-left interface

The application changes both language and document direction.

Real RTL behavior

WHAT THE SCREEN PROVES

Arabic is not only translated: navigation, hierarchy, alignment and reading flow are mirrored for right-to-left use.

- Document-level RTL direction
- Arabic typography and translated interface copy
- Layout-specific alignment corrections
- Latin digits retained for technical readability

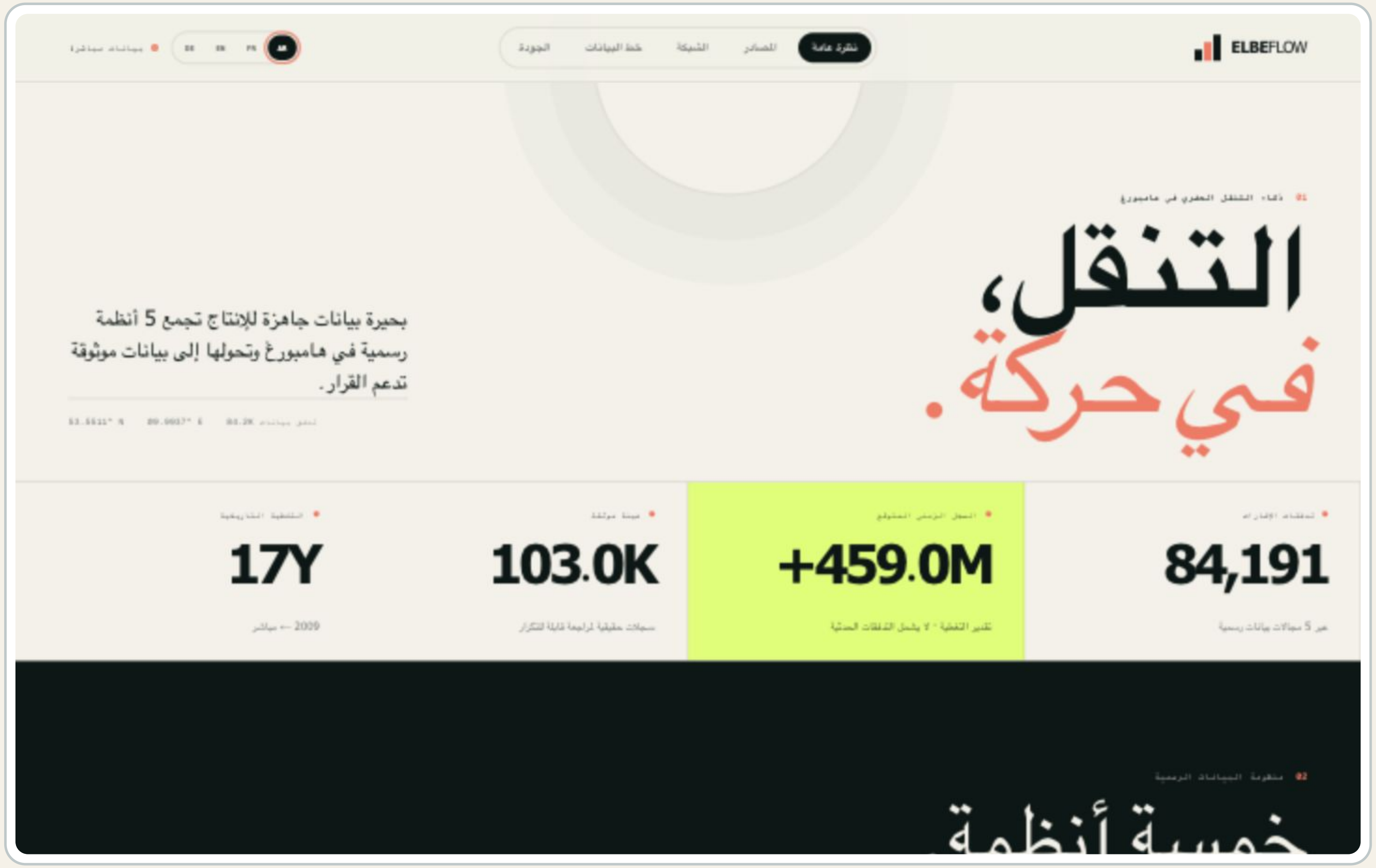

Figure 8 — Arabic overview with genuine RTL layout behavior.

Mobile experience

The same decision product remains usable on a 390 × 844 viewport.

RESPONSIVE PRODUCT

Designed beyond the desktop demo

The mobile layout preserves the live state, core scale metrics and language controls without horizontal overflow. Dense analytical sections collapse into a readable single-column experience.

- Verified at 390 × 844 pixels
- No horizontal document overflow
- Accessible language controls remain available
- Headline metrics preserve their visual priority

Figure 9 — German mobile overview captured from the running application.